Contact

lee.trent.1@gmail.com

www.linkedin.com/in/leetrent (LinkedIn)

Top Skills

Large Language Models (LLM)

Retrieval-Augmented Generation (RAG)

QLoRA Fine-Tuning

Certifications

AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents

Lee Trent

Software Engineer

Kansas City, Missouri, United States

Summary

STRATEGIC SOFTWARE ENGINEER | AI ENGINEER
I am a veteran software engineer with 30 years of experience designing and deploying complex, mission-critical systems for federal agencies, publicly traded corporations, and private firms. My career has been defined by a commitment to data quality, architectural integrity, and the evolution of modern enterprise stacks—from Java and .NET to Go and now, production-ready Artificial Intelligence.

Through my consultancy, TrentTEK LLC, I bridge the gap between "experimental" AI scripts and robust, enterprise-grade software. By combining decades of full-stack architecture experience—spanning relational databases (Oracle, SQL Server, MySQL) and sophisticated front-end interfaces—with advanced engineering in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic workflows, I build intelligent solutions designed for scalability, security, and accuracy.

I bring the discipline of 30 years in the industry—mentoring teams, capturing complex business rules, and leading end-to-end design—to the frontiers of modern AI automation.

TECHNICAL FOUNDATION:
AI ENGINEERING: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), QLoRA Fine-Tuning, and Autonomous Agents.
SOLUTION STACKS: .NET and Spring Framework.
DATA ARCHITECTURE: Expert-level proficiency in Oracle, MySQL, and MS SQL Server.
LEADERSHIP: Former Application Architect and Team Lead for large-scale federal and commercial projects.

Experience

TrentTEK LLC

Owner and Software Engineer

September 2021 - Present (5 years 2 months)

Kansas City Metropolitan Area

TrentTEK LLC is an enterprise software engineering consultancy that previously provided specialized development for the U.S. General Services Administration (GSA), utilizing the Microsoft .NET solution stack and MySQL to architect complex human resources data quality systems.

Following the successful completion of our federal contracting commitments, the firm executed a strategic pivot into AI Engineering. To establish a production-grade foundation, we completed comprehensive technical deep dives into the AI Engineer Core Track (covering LLM Engineering, advanced RAG architectures, QLoRA fine-tuning, and LangChain/agentic frameworks).

TrentTEK now leverages decades of enterprise data architecture experience to design, test, and implement secure, deterministic AI systems and autonomous agents built for corporate and agency scale.

Fluent Consultants

Senior Software Developer

January 2021 - September 2021 (9 months)

Kansas City, Missouri, United States

Member of a development team that developed a new laboratory information management system for MRI Global (https://www.mriglobal.org).

Technologies used on this project:
ASP.NET 5.0 Razor Pages, Entity Framework Core 5.0, Microsoft SQL Server, Telerik's Kendo UI for jQuery.

Holland Technologies

Application Architect / Software Engineer

August 2009 - November 2020 (11 years 4 months)

Kansas City Metropolitan Area

Consulted at the U.S General Services Administration (GSA) where I played a key role in the development of the following web applications:

Human Resources Quality Management System (HR-QMS):
HR-QMS is a web application developed to support GSA’s operational improvement initiative, focusing on HR and Payroll data quality.

Partnered with the application architect on this project to develop this web application using the following technologies: ASP.NET Core MVC, Javascript, SecureAuth Single Sign-On, Entity Framework, MySQL DBMS.

Learning Academy User Registration Site (LA-URS):
Lead the end-to-end design and development effort of the GSA Learning Academy User Registration Site, a GSA owned solution that allows external customers to register and request enrollment in programs offered by the GSA Learning Academy.

This public facing web application was developed using ASP.NET Core Razor Pages, Identity Framework and Entity Framework.

Identity Framework was used for user authentication and role-based authorization. Users are required to confirm their email addresses and enable 2-factor authentication using a smartphone authentication application to complete the registration process. User login requires a username, password and an authenticator verification code.

Entity Framework was used for MySQL DBMS interaction.

Volunteer Leave Transfer Program (VLTP):
Developed the multi-factor authentication (MFA) component for this public facing web application using ASP.NET Core MVC and MAX.gov as the external identity provider.

Authorized Leave and Overtime Help Application (ALOHA):
Lead the end-to-end design and development effort of GSA's Authorized Leave and Overtime Help Application, a system that served the entire agency from 2012 to 2018.

This web application was developed using Java Server Faces (JSF), ICEfaces, Javascript, Enterprise Java Beans (EJB), Java Persistence API (JPA), Single Sign-On (SecureAuth) and Oracle DBMS.

Booz Allen Hamilton
Application Developer
March 2009 - August 2009 (6 months)
Kansas City Metropolitan Area

Consulted at the U.S Department of Agriculture (USDA) - Farm Service Agency (FSA) as an application developer.

Played a key developer role in the redesign and development of FSA's Farm Records software to conform to FSA's new architectural specification

My role with Booz Allen Hamilton also included making design recommendations, mentoring junior members of the development team and interviewing applicants for developer roles on other USDA projects awarded to Booz Allen Hamilton.

Spring Framework 2.0 was used for its inversion of control (dependency injection), aspect-oriented programming (Spring AOP) and data access capabilities. Spring AOP was used to handle logging and exception handling in both the web and middle tiers. Spring Dependency Injection was used to inject business services into Struts Action classes and to inject data access objects into the business services. Spring JDBC was used to perform all CRUD (create, retrieve, update, delete) operations on the Farm Records database. The Farm Records web service was rewritten using Apache's Axis 1.4.

The Scrum methodology of agile software development was used on this project.

FSA's Farm Records was running at the time on a production platform consisting of Websphere Application Server 6.1 and MS SQL Server 2005 DBMS.

Development platform included Eclipse 3.4, JBoss 4.2.2 and MS SQL Server 2005. Maven was used to build the project which consisted of the following Farm Records components: web user interface, web service, business services, business domain and shared utilities.

Tek Systems
Application Developer
November 2008 - January 2009 (3 months)
Overland Park, KS

Consulted at Secure Passage (later renamed FireMon) as an application developer.

Secure Passage at the time was an industry-leading independent software vendor focused on developing security applications. Their flagship product, FireMon, monitors critical network devices so administrators can see, audit and manage their networks.

Hired on as a contractor to help two other developers convert the FireMon desktop client application (written in C#) to a Java-based web application running on Tomcat

Tasks included extracting business rules and requirements from the legacy code base and making design recommendations to assure that these business rules were included in the new web application.

Spring Framework 2.0, Apache Struts 2.0, OpenSymphony 's SiteMesh, the Prototype Javascript Framework, HTML and CSS were used to develop the web tier and the Apache CXF open source services framework was used to interact with FireMon's proprietary application server.

Project was put on hold due to the company’s abrupt reorganization and was eventually cancelled.

RiverPoint Group LLC
Application Developer
October 2007 - October 2008 (1 year 1 month)
Lenexa, KS

Consulted at Silpada Designs, Inc. as an application developer. Silpada Designs at the time was the largest direct seller of jewelry to consumers in the United States.

Rewrote Silpada's public website using Java Server Faces and Facelets technologies.

Because of Silpada's rapid growth and the scalability challenges exposed in the current vendor supplied system, Silpada decided to develop its own order entry system. Was solely responsible for the design and development of the middle and persistent tier components of Silpada’s new order entry system, a system that at the time was used by approximately 30,000 representatives in the United States and Canada selling directly to the public.

Worked closely with the systems analyst on a daily basis to capture all business rules associated with this complex order entry system to assure that all functionality provided by the legacy system was captured in the new system. Design patterns used: Strategy, Template Method, Factory, Façade, Transfer Object and Service Locator.

JSF/Facelets, EJB 3.0 and Java Persistence API (JPA) were used to develop the web tier, middle tier and persistence tiers of this new order entry system, respectively. Hibernate 3.2 served as the persistence provider for JPA. JDK 1.5 was used throughout. All middle tier components were tested using JUnit.

All of Silpada's Java applications ran on a production platform consisting of JBoss Application Server, Ubuntu Linux Operating System and Microsoft SQL Server 2005. MyEclipse running on Windows XP was used as the IDE.

Intronic Solutions Group

Application Developer

July 2006 - October 2007 (1 year 4 months)

Overland Park, KS

Consulted at Sprint Nextel as an application developer.

Member of the original Digital Lounge design and development team. This project was a complete rewrite of the existing digital media content purchasing system.

During this time, visitors to Sprint’s Digital Lounge were able to easily find, preview, buy and manage digital media content such as ringers, screensavers, games, applications, multimedia themes, music and call tones. The Digital Lounge provided online access to Sprint’s full portfolio of digital media content available on Sprint devices. Digital media content was updated weekly with releases of new and popular content, as well as special promotions.

Was solely responsible for the design and development of the following web components: "Purchase", "Refund", "Unsubscribe", "Give-a-Gift", "Tell-a-Friend", "Recently Viewed Products" and “My Content Manager”. The initial release of Digital Lounge was on 11/12/06 and was followed by many subsequent releases.

Wrote detailed design documents in response to use cases written by the systems analyst. This document included UML class and sequence diagrams from which the components described above were built.

Struts, Tiles, Java Server Pages (JSPs), Tag Libraries (JSTL, Struts and Custom), AJAX, JavaScript, and HTML/CSS were used to develop Digital Lounge as a portlet to run on Sprint's
Websphere Portal Server (WPS) / Websphere Application Server (WAS) / Advanced Interactive eXecutive (AIX) platform. Websphere Studio Application Developer (WSAD) running on Windows XP was used as the Integrated Development Environment (IDE).

NAIC
Application Developer
February 1997 - June 2006 (9 years 5 months)
Kansas City Metropolitan Area

System for Electronic Rate and Form Filing (SERFF):
SERFF a system used in all 50 states by over 5,000 insurance companies for the purpose of obtaining state regulatory approval of new insurance products and changes in insurance rates.

Member of the development team responsible for the migration of the SERFF application from a Domino/Lotus Notes platform to NAIC's HP-UX/JRun/Oracle platform using Java technologies.
Wrote design approach documents in response to use cases written by business analysts. This document included UML class diagrams and data modeling diagrams from which Java classes and Oracle tables were created.

JDBC data access objects mapped to Oracle tables handled data persistence and model objects where used to enforce business rules. The user interface components were written using Struts, Tiles, Java Server Pages, Java Standard Tag Libraries (JSTL), JavaScript and HTML/XHTML/CSS. To interface with SERFF's state and industry business partners, two web services were developed using JAX-RPC technology.

All applications were deployed as web archives (WAR files) and were built using Ant technology All components were unit tested using JUnit.

State Based Systems (SBS):

SBS provides a suite of insurance regulatory software for member states. Modules include: Agent and Company Licensing, Continuing Education, Consumer Complaints, Regulatory Actions, Fraud and Enforcement.

Lead a development team responsible for the design and development of the SBS Consumer Complaints System. Solely responsible for capturing user requirements, data and class modeling of the new system and making all architectural decisions to assure that this newly developed system would meet all functional and non-functional requirements and integrate cleanly with SBS and other NAIC systems.

UML, Data Modeling, Java Technology (Servlets/JSPs/JDBC), XML, SQL, HTML/CSS and JavaScript were used to design and develop these applications to run on a SunOne/Windows/Oracle platform.

YRC Freight
Programmer Analyst
May 1995 - February 1997 (1 year 10 months)
Overland Park, KS

Member of a large development team led by Andersen Consulting that migrated Yellow Freight's freight hauling applications from a mainframe environment to a three tier client-sever platform consisting of:
- Oracle DBMS
- Business logic coded in C on a UNIX platform (Sun OS)
- GUI written in PowerBuilder for a Windows client

Designed, coded and tested programs written in the C programming language for the following freight hauling applications:
- Activity Based Cost Modeling
- Freight Flow Management (Computer Assisted Dispatch)
- Bill of Lading Messaging between Legacy and UNIX platform

All C Programs included embedded SQL to interact with Oracle DBMS and made extensive use of Oracle host arrays to improve performance. Batch programs employing checkpoint, restart/recovery processing were used to manage large volumes of data.

Education

State University of New York at Stony Brook
Bachelor of Arts

Johnson County Community College
Information Technology